Mathematical Modeling

Lab 2: distributions and courier deliveries

Personal study notes

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1 From a statement to solver matrices

Both tasks ask us to choose quantities subject to linear restrictions. First define the variables, derive the objective and constraints, and check feasibility. Only then assemble the numerical arrays. We use column decision vectors and the convention

$$\min f^T v, \qquad Av \le b, \qquad A_{\text{eq}} v = b_{\text{eq}}.$$

Every row is one restriction; every column belongs to one decision variable. Bounds can be written as matrix rows:

$$\ell \le v \le u \quad \Longleftrightarrow \quad \underbrace{\begin{bmatrix} -I \\ I \end{bmatrix}}_{A} v \le \underbrace{\begin{bmatrix} -\ell \\ u \end{bmatrix}}_{b}.$$

The lower bound changes sign because $v_i \ge \ell_i$ becomes $-v_i \le -\ell_i$. The MATLAB scripts below put all bounds into these rows and pass empty lower/upper-bound arguments to `linprog`. Using its `lb`, `ub` arguments instead describes the same feasible set.

2 Group 1: choosing a probability distribution

2.1 Task statement

Dana jest zmienne losowe X oraz Y o dziedzinie $[5] = \{1, 2, 3, 4, 5\}$ przyjmujące wartości zgodnie z macierzami przedstawionymi na dole tego pliku. Naszym celem jest znalezienie rozkładu prawdopodobieństwa $P = (p_i)_{i \in [5]}$ takiego, który minimalizuje wartość:

a) $E_P[X]$ (przy dodatkowym założeniu, że każde prawdopodobieństwo p_i (dla $i \in [5]$) jest co najwyżej równe 0.3),

b) z ograniczającą z góry jednocześnie wartości: $E_P[x]$ oraz $E_P[Y]$.

Zaprojektuj i zaimplementuj dwa modele programowania liniowego, które znajdą optymalne rozkłady prawdopodobieństwa dla powyższych podpunktów. Ile wynoszą wartości oczekiwane w każdym z tych przypadków? W każdym podpunkcie, zwizualizuj uzyskane rozkłady prawdopodobieństwa przy pomocy wykresów słupkowych (przydatna może być funkcja `bar`). The supplied value arrays are

i	1	2	3	4	5
$X(i)$	1.2	2.1	-0.4	1.2	-0.5
$Y(i)$	1.2	-0.6	2.4	-0.1	0.5

.

2.2 Variables and expectations

The outcomes and their values are fixed. The decisions are the probabilities $p = (p_1, \ldots, p_5)^T$, with $p_i \geq 0$ and $\sum_i p_i = 1$. Use the value vectors

$$\xi = (1.2, 2.1, -0.4, 1.2, -0.5)^T, \qquad \eta = (1.2, -0.6, 2.4, -0.1, 0.5)^T.$$

Then

$$E_P[X] = \xi^T p, \qquad E_P[Y] = \eta^T p.$$

These are weighted averages, so they are linear in the unknown probabilities. The 0.3 cap is an extra condition in part (a) only.

2.3 Part (a): minimize one expectation

The complete LP is

$$\min \xi^T p, \qquad \begin{bmatrix} -I_5 \\ I_5 \end{bmatrix} p \leq \begin{bmatrix} 0_5 \\ 0.3\,\mathbf{1}_5 \end{bmatrix}, \qquad \mathbf{1}_5^T p = 1.$$

To lower the expectation, place as much probability as allowed on the smallest values: $X(5) = -0.5$ and $X(3) = -0.4$ each receive 0.3. The remaining 0.4 goes to outcomes 1 and 4, both with value 1.2. Outcome 2 has a larger value and receives zero. Thus all optimal distributions are

$$p = (t, 0, 0.3, 0.4 - t, 0.3)^T, \qquad 0.1 \leq t \leq 0.3.$$

The interval ensures that both p_1 and p_4 remain between zero and 0.3. For a direct lower-bound certificate, eliminate $p_1 + p_4$ using the sum:

$$E_P[X] = 1.2 + 0.9p_2 - 1.6p_3 - 1.7p_5 \geq 1.2 - 1.6(0.3) - 1.7(0.3) = 0.21.$$

Every displayed distribution attains this bound. Its other expectation is

$$E_P[Y] = 0.83 + 1.3t \in [0.96, 1.22].$$

For example, $p = (0.1, 0, 0.3, 0.3, 0.3)^T$ gives $E_P[X] = 0.21$ and $E_P[Y] = 0.96$. Different optimal bar charts can therefore be equally correct: part (a) minimizes $E_P[X]$, not $E_P[Y]$.

2.4 Part (b): minimize a common upper bound

For a fixed distribution, the smallest number bounding both expectations is $\max\{E_P[X], E_P[Y]\}$. Introduce z as a sixth decision variable and solve

$$\min z, \qquad \xi^T p - z \le 0, \qquad \eta^T p - z \le 0, \qquad p \ge 0, \quad \mathbf{1}_5^T p = 1.$$

For $v = (p_1, \dots, p_5, z)^T$, the solver arrays are

$$f = \begin{bmatrix} 0_5 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} \xi^T & -1 \\ \eta^T & -1 \\ -I_5 & 0_5 \end{bmatrix}, \quad b = 0_7, \quad A_{\text{eq}} = \begin{bmatrix} \mathbf{1}_5^T & 0 \end{bmatrix}, \quad b_{\text{eq}} = 1.$$

Here A has seven rows and six columns. Its last five rows restrict the probabilities, leaving z unrestricted in sign. The 0.3 cap is absent.

Hand solution and a global certificate. Outcome 5 favors X; outcome 2 favors Y. Mixing just these two, set $p_2 = t$, $p_5 = 1 - t$. Their expectations are

$$E_P[X] = -0.5 + 2.6t, \qquad E_P[Y] = 0.5 - 1.1t.$$

Balancing them gives $3.7t = 1$, hence

$$p^* = (0, 10/37, 0, 0, 27/37)^T, \qquad E_{P^*}[X] = E_{P^*}[Y] = z^* = 15/74 \approx 0.202703.$$

Balancing a chosen pair alone would not rule out a better mixture involving other outcomes. To prove global optimality, observe that

$$\frac{11\xi + 26\eta}{37} = \left(\frac{6}{5}, \frac{15}{74}, \frac{58}{37}, \frac{53}{185}, \frac{15}{74}\right)^T \ge \frac{15}{74}\,\mathbf{1}_5.$$

For any feasible distribution and upper bound,

$$z \ge \frac{11}{37} E_P[X] + \frac{26}{37} E_P[Y] \ge \frac{15}{74} \sum_i p_i = \frac{15}{74}.$$

Our distribution attains this bound. Equality requires zero mass at outcomes 1, 3 and 4 and equal expectations, so this optimum is unique.

2.5 MATLAB implementation and comparison

Run `lab_02_probability.m` from the repository root:

```
run('matlab/labs/lab_02_probability.m')
```

The script prints both expectations and the probability vectors, draws one bar chart for each part, and checks feasibility and objective values with absolute tolerance 10^{-7}. It uses solver constraint and optimality tolerances 10^{-9}. For part (a), it checks the optimal family rather than requiring one particular vector. For part (b), it checks the unique vector above.

3 Group 5: deliveries along a fixed courier route

3.1 Task statement

W pewnej firmie kurierskiej kurier przewozi paczki. Swoją trasę rozpoczyna w centrum logistycznym, a następnie podąża zgodnie z trasą przedstawioną na poniższym rysunku (centrum logistyczne oznaczono jako 0), kończąc w punkcie 0 po odwiedzeniu wcześniej wszystkich punktów.

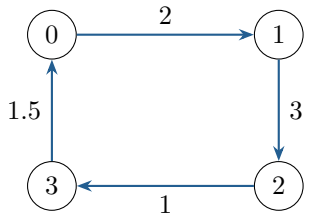


Figure 1: The prescribed route and its edge cost weights. Only delivery quantities are decisions; the route is fixed.

Kurier rozpoczyna podróż z ładunkiem M kilogramów. Następnie, w każdym kolejnym punkcie podróży zostawia pewną liczbę kilogramów, przy czym:

- w punkcie 1 musi zostawić przynajmniej 2 kilogramy, a maksymalnie 6 kilogramów,
- w punkcie 2 musi zostawić przynajmniej 4 kilogramy, a zostawić maksymalnie 7 kilogramów,
- w punkcie 3 musi zostawić przynajmniej 2.5 kilograma, a zostawić maksymalnie 5 kilogramów.

Zaprojektuj odpowiedni model programowania liniowego, który wyznaczy, ile kilogramów musi zostawić w każdym z punktów tak, aby spełnić warunki i zminimalizować koszt podróży, który liczony jest następująco:

- koszt przejechania krawędzi to jej waga przemnożona przez wagę wiezionego ładunku,
- koszt całej podróży to sumaryczny koszt przejechanych krawędzi.

Rozwiąż zaprojektowany model przy pomocy funkcji `linprog` kolejno dla wartości $M = 7, 8, 9, 10, 11$. Ponadto, przy pomocy wykresu słupkowego zwizualizuj uzyskane rozwiązania dla każdego punktu na trasie i każdej wartości M (przydatna może być funkcja `bar`).

3.2 Variables, loads, and the objective

Let x_i be kilograms delivered at point i. The initial mass M is a given parameter for each run. Masses are continuous. The load decreases after each delivery, so the objective is the sum of the following costs:

Edge	Load on departure (kg)	Edge cost
$0 \to 1$	M	$2M$
$1 \to 2$	$M - x_1$	$3(M - x_1)$
$2 \to 3$	$M - x_1 - x_2$	$M - x_1 - x_2$
$3 \to 0$	$M - x_1 - x_2 - x_3$	$1.5(M - x_1 - x_2 - x_3)$

Thus the total trip cost is

$$C_M(x) = 7.5M - 5.5x_1 - 2.5x_2 - 1.5x_3.$$

Delivering one kilogram at point 1 saves the weights of all remaining edges, $3 + 1 + 1.5 = 5.5$. The analogous savings at points 2 and 3 are 2.5 and 1.5. This explains the negative coefficients. The source does not specify a currency unit, so costs are reported in the units induced by its edge weights.

3.3 The complete matrix model

With $x = (x_1, x_2, x_3)^T$, minimize $f^T x$ using

$$f = \begin{bmatrix} -5.5 \\ -2.5 \\ -1.5 \end{bmatrix}, \qquad \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{A \in \mathbb{R}^{7\times 3}} x \leq \underbrace{\begin{bmatrix} M \\ -2 \\ -4 \\ -2.5 \\ 6 \\ 7 \\ 5 \end{bmatrix}}_{b}.$$

There are no equality constraints. The first row prevents delivery of more than the initial load. Because all deliveries are nonnegative, this also ensures that every earlier edge load is nonnegative. The next three rows are the minimum deliveries; the last three are the maximum deliveries. The condition $x_1 + x_2 + x_3 \geq 0$ is already implied and needs no extra row. The statement allows returning with undelivered load; an equality $\sum_i x_i = M$ need not be imposed as a modeling assumption.

The term $7.5M$ is constant within a run, so omitting it from the solver objective preserves the optimizer. Add it back to the returned objective value when reporting the cost of the trip, particularly when comparing runs with different M.

3.4 Hand solution and feasibility

The minimum deliveries total $2 + 4 + 2.5 = 8.5$ kg. Thus $M = 7$ and $M = 8$ are infeasible. For $M = 9, 10, 11$, start with these minimum deliveries and allocate the remaining $M - 8.5$ kg. Each extra kilogram saves most when delivered at point 1. Its upper bound allows four extra kilograms, more than the 0.5, 1.5, or 2.5 kg available in these cases. Hence

$$x^*(M) = (M - 6.5, 4, 2.5)^T, \qquad M \in \{9, 10, 11\}.$$

For a global certificate, write $S = x_1 + x_2 + x_3$. Every feasible point satisfies

$$5.5x_1 + 2.5x_2 + 1.5x_3 = 5.5S - 3x_2 - 4x_3 \leq 5.5M - 3(4) - 4(2.5) = 5.5M - 22.$$

Consequently $C_M(x) \geq 2M + 22$, with equality at the stated delivery vector. Equality requires $S = M$, $x_2 = 4$, and $x_3 = 2.5$, so each feasible case here has a unique optimizer.

M (kg)	x_1 (kg)	x_2 (kg)	x_3 (kg)	Minimum trip cost
7			Infeasible	
8			Infeasible	
9	2.5	4	2.5	40
10	3.5	4	2.5	42
11	4.5	4	2.5	44

All available load is delivered at these optima because each delivery saves cost and the upper bounds do not prevent it. This is a conclusion for these inputs, rather than an extra equality added to the task.

3.5 MATLAB implementation and comparison

Run `lab_02_courier.m`:

```
run('matlab/labs/lab_02_courier.m')
```

The script runs all five requested loads, checks infeasibility for 7 and 8, and compares the remaining cases with the hand solution. It independently recomputes the full trip cost from the four edge loads. Solver tolerances are 10^{-9}; comparison tolerance is 10^{-7}. The grouped bar chart retains ticks for every M and uses missing values for infeasible cases: zero bars would incorrectly suggest a feasible zero-delivery solution.

4 Sources and editorial remarks

Task sources: Hubert Grochowski, `Sam2Gr1.pdf` and `Sam2Gr5.pdf`, one page each, headed Lab 2, academic year 2026/2027. Originals are at `../Labs/tasks/` relative to the repository root. The mathematical task wording is retained in Polish above, with the supplied arrays typeset as a table and the courier diagram redrawn. Administrative submission instructions are omitted. The lowercase x in the probability statement's $E_P[x]$ is interpreted as the named random variable X.

The user's completed courier submission is `../Labs/Lab2/RadoslawGlasekLab2.m`; the user reports full lab credit. `../Labs/Lab2/Lab2.m` is a friend's probability solution and was consulted as a comparison, not copied into the companions. The Lab 3 file is a starter copy, not evidence of completed Lab 3 work. The original files are preserved. The matrix-only formulations, hand certificates and companion scripts are independent study additions.

Exact rational checks confirm the displayed supporting coefficients, minimax mixture and courier costs. The agent attempted to run both companions through the installed MATLAB launcher on 25 October 2026; it exited with code 1 and no diagnostic before producing output. Actual MATLAB execution and plot rendering therefore remain unverified. Expected results and acceptance tolerances are given above; execution status is also recorded in `docs/matlab-verification.md`.